

FLORAS-AI

Urban Canopy Intelligence Platform

Empowering communities to track, verify, and preserve the urban canopy through collective citizen intelligence and AI diagnostics.

THE PROBLEM: COLLAPSING CANOPIES

Our Cities Are Losing Their Vital Forests

- **Urban Canopy Decline:** Cities lose millions of trees annually to construction, pests, and climate stress, exacerbating the dangerous "urban heat island" effect.
- **Delayed Municipal Action:** Local park authorities and municipal arborists operate on long, manual inspection cycles. Sick or diseased specimens go unnoticed until it is too late.
- **Segmented Eco-Data:** There is no unified, geo-tagged registry for citizens to report tree concerns or track local reforestation efforts.
- **Low Stewardship Engagement:** Citizens care deeply about their local ecosystems but lack reward-driven, simple channels to participate in civic conservation.

THE SOLUTION: CITIZEN SCIENCE & AI

Floras-AI Collaborative Ecosystem

- **Live Geospatial Mapping:** A mobile-first interactive map allowing volunteers to catalog trees, submit photos, and report structural/health changes in real-time.
- **Instant AI Diagnostics:** The "Tree Doctor" chat widget provides immediate, arborist-guided triage to assess tree symptoms and advise whether to flag them as "sick".
- **Double-Blind Verification:** Independent citizen scientists validate reported trees, adding a verified checkmark and map ring when consensus is met.
- **Actionable Authority Dispatch:** Seamless compiler tools that generate official, filtered PDF summaries and CSV spreadsheets for local municipal maintenance dispatch.

PRODUCT FEATURES

A Unified Environmental Patrol Platform

Module	Function	Outcome
Interactive Map	Geotagging, status tracking, Leaflet GIS layers	Real-time urban canopy database
Tree Doctor AI	Groq Llama-3.3-70b-versatile secure endpoint	Instant arborist symptoms diagnostics
Consensus Verification	Citizen double-confirmation logic & RLS profiles	Pre-validated data preventing spam
Guardian Leaderboard	Gamified engagement points & badges	High repeat citizen contribution rate
Authority Report	One-click PDF/CSV neighborhood filter compiler	Fast coordination with city arborists

Secure & Modular Application Flow

1. User Interface (React + Vite + Leaflet) captures geo-stamped registry requests.
2. Chat requests proxy through secure serverless functions (/api/chat) on Vercel/Netlify.
3. Groq completions API processes data under llama-3.3-70b-versatile with custom prompts.
4. Supabase DB handles storage of high-res photos, RLS authentication, and profiling.
5. Client compiles PDF reports and canvas card downloads locally for offline performance.



The Citizen Forester Lifecycle

- Log Specimen: Volunteer snaps a photo of a tree. AI auto-identifies species or logs it manually.
- Symptom Query: Volunteer spots leaf discoloration. Ask AI Doctor opens pre-loaded with tree context.
- Diagnoses Triage: Llama-3.3 determines the issue (e.g. fungal rust), and triggers a "Sick" flag in the registry.
- Community Consensus: Other local users view the map pin, confirm symptoms in-person, triggering "Verified" status.
- Civic Action: Neighborhood groups export the weekly verified sick list to the municipal arborist for pruning/care.

Stakeholders & Target Audience

- **Municipal Authorities:** City councils and arborist departments looking for low-cost, crowd-sourced canopy health analytics and immediate emergency hazard logs (fallen branches, disease outbreak).
- **Environmental NGOs & Forestry Initiatives:** Organizations managing city greening programs that need structural tooling to coordinate neighborhood planting, adoptions, and tree survival tracking.
- **Active Communities & Citizens:** Neighborhood associations, schools, and green volunteers looking for actionable, gamified tools to monitor and improve the local urban environment.

Ensuring Longevity and Platforms Expansion

- Open-Source Collaboration: Free core mapping tools to maximize community adoption and developer contributions.
- SaaS Council Dashboard: A paid subscription portal for city councils integrating direct arborist dispatches, work order automation, and heat map monitoring.
- Corporate Social Responsibility (CSR): Partnering with corporations to sponsor urban reforestation and local adoption drives, boosting brand visibility.
- Scientific Datasets: Providing anonymized canopy development and disease vectors data to university researchers and urban climate planners.

Empowering Citizen Guardians for Greener Cities

- Climate Action: Faster disease response saves mature urban trees. A single mature tree absorbs up to 48 lbs of CO2 and produces enough oxygen for 2 people daily.
- Environmental Literacy: AI diagnostics teach volunteers about local biodiversity and arborist symptoms.
- Social Cohesion: Gamified leaderboards and adoptions foster neighborhood pride and clean, green community spaces.

Join the Canopy Movement today.

Deploy at github.com/Manindra-babu/Floras-AI